

Poster Session Synthesis

EWGT 2026 — Session 2P, Day 2 (Thursday, 3 September 2026)

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Overview

- Session 2P: 21 posters.
- 15:30
- Across the posters, three concerns recur:
 - how to **measure** transport systems more precisely,
 - how to **operate** them more efficiently, and how to make them more
 - **sustainable, resilient, and equitable.**
- 6 themes

Themes

- 1 Safety, equity, and economics
- 2 Freight, logistics, and last-mile delivery
- 3 Fleet electrification and energy
- 4 Digital platforms, data, and AI tools
- 5 Reliability and resilience under disruption
- 6 Traffic state and flow estimation

2. Freight, logistics, and last-mile delivery (1/2)

- **P4.** *On Assessing the Performances of Freight Delivery Operated by Passenger Trains: A Discrete Event Simulation Approach*

Summair Anis, Amirhossein Kalantari, Nicola Sacco (University of Genoa, IT); Valentina Moretti (Hitachi Rail STS, IT)

- **P5.** *Two-Phase Optimization Framework for Truck Shipment Consolidation at a Cross-Dock Facility in the Philippines*

Rovie Manansala, Francis Adrian Viernes, Varsolo Sunio (University of Asia and the Pacific, PH)

- **P8.** *Comparing direct-to-customer delivery and microhub-based cargo bike delivery: a COPERT-based tailpipe emissions assessment for B2B logistics in Rome*

Siavash Pourkhosro, Marzieh Afsari, Lory Michelle Bresciani Miristice, Guido Gentile (Sapienza University of Rome, IT)

Synthesis (part 1)

- P4 assesses freight delivery performance operated by passenger trains, via discrete event simulation.
- P5 optimizes truck shipment consolidation at a cross-dock facility, via a two-phase framework.
- P8 compares tailpipe emissions of direct-to-customer versus microhub cargo bike delivery, using COPERT.

2. Freight, logistics, and last-mile delivery (2/2)

- **P16.** *Intermodal Inland Terminal Design Methodology - Current Status and Future Directions*
Faustyna Finke, Piotr Sawicki (Poznan University of Technology, PL)
- **P19.** *Enhancement of selected routing heuristics efficiency by pre-clustering of requests for deliveries*
Oleksandr Yakymenko, Mateusz Turek, Vitalii Naumov (Cracow University of Technology, PL)
- **P21.** *A Two-Echelon Routing Problem for Last-Mile Delivery in Mountainous Regions Using a Combination of Buses and Unmanned Aerial Vehicles*
Nicola Sacco, Aliakbar Shahabi, Alireza Shahedi, Angela Di Febbraro (University of Genoa, IT)

Synthesis (part 2)

- Largest theme in the session by poster count (6 posters across parts 1 and 2).
- P16 reviews current status and future directions for intermodal inland terminal design methodology.
- P19 and P21 both address routing problems: heuristic efficiency via pre-clustering (P19), and a two-echelon bus/UAV routing problem for mountainous last-mile delivery (P21).

1. Safety, equity, and economics (1/2)

- **P2.** *Mobility Experiences for Future Roads: A Cross-Regional Citizen Perspective on Integrated, Autonomous, and Inclusive Transport*

Hazem Ali (Halmstad University, SE); Indhumathi Ravichandran (SRM University, IN); Radha Reddy (Manipal University, IN); Harrison Kurunathan (CISTER, PT); David Pereira (Vortex Colab, PT)

- **P11.** *Integration of safety into a comprehensive sustainability indicator for small-medium Ports*

Margarida Coelho (University of Aveiro, PT); Flavia Ferretti, Federico Orsini, Riccardo Rossi (University of Padua, IT); Maria Manuel Cruz (APA – Administration of the Port of Aveiro, PT)

Synthesis (part 1)

- P11 integrates safety into a sustainability indicator for small-medium ports.
- P2 presents a cross-regional citizen perspective on integrated, autonomous, and inclusive transport.

1. Safety, equity, and economics (2/2)

- **P14.** *Multidimensional transport opportunity cost assessment: Evidence from the European Union and Ukraine*

Krzysztof Wyrwicki, Adam Redmer (Poznan University of Technology, PL); Mariya Khmelyarchuk (Ivan Franko National University of Lviv, UA)

- **P15.** *Definition of safety baseline for road safety assessments on regional road networks*

Tevoh Lordswill Ndingwan, Isaac Ndumbe Jackai II, Elvis Chia Ngwah, Luca Persia (Sapienza University of Rome, IT); Bryan Gonzalez Hernandez (University of Napoli Federico II, IT)

Synthesis (part 2)

- P15 defines a safety baseline for road safety assessments on regional road networks.
- P14 assesses multidimensional transport opportunity cost, with evidence from the EU and Ukraine.

3. Fleet electrification and energy

- **P12.** *A Process-Modular Facility Location Model for the Design of Sustainable Aviation Fuel Supply Chains*

Gaétan Richoux, Guillaume Martin, Didier Gourc (IMT Mines Albi, Centre Génie Industriel, FR); Mariam Lafkihi, Shenle Pan (Mines Paris - PSL, Centre for management science CGS, FR)

- **P13.** *A cluster-based spatial planning model for charging distribution in urban areas*

Aleksandra Colovic, Mario Marinelli, Michele Ottomanelli (Polytechnic University of Bari, IT)

- **P20.** *Estimating Electric Bus Energy Consumption from Real-World Operational Data: A Comparison of Data-Driven and Physics-Based Approaches*

Vincenza Torrisi, Pierfrancesco Leonardi, Stefania Vitale, Matteo Ignaccolo (University of Catania, IT)

Synthesis

- Three scales of the energy transition: supply chain design for sustainable aviation fuel (P12), urban charging distribution planning (P13), and electric bus energy consumption modeling (P20).
- P20 compares data-driven and physics-based modeling approaches, using real-world operational data.

4. Digital platforms, data, and AI tools

- **P7.** *Integrating Public Transport Data in a Data Space: The i-MovE Mobility Platform*
Marc Manzanares, Rubén Palà (Universitat Politècnica de Catalunya, ES)
- **P10.** *Intelligent Mobility Space – a city quarter as a living lab, echoed in a digital work-life-twin*
Andreas Wendemuth, Leander Kauschke (Otto-von-Guericke University Magdeburg, DE)
- **P17.** *A Hybrid Rule-Based and LLM Pipeline for Natural Language Querying of Live Bikeshare Station Data: A Serverless DuckDB-WASM Implementation*
Mehmet Metin Kunt (Eastern Mediterranean University, TR)

Synthesis

- P7 integrates public transport data in a shared data space (the i-MovE platform).
- P10 represents a city quarter, run as a living lab, through a digital work-life-twin.
- P17 enables natural language querying of live bikeshare data, via a hybrid rule-based and LLM pipeline.

5. Reliability and resilience under disruption

- **P6.** *GAN-based synthetic data augmentation for improved stability of railway delay predictions*

Mick Molitor, David Rößler-von Saß, Natalia Kliwer (Freie Universität Berlin, DE)

- **P9.** *Demand-Integrated Resilience Assessment of a Circular Electric Bus Network with Entropy-Based Weighting*

Vishikha Keshava, Raviraj H Mulangi (National Institute of Technology Karnataka, IN); Anila Cyril (National Transportation Planning and Research Centre, IN)

- **P18.** *Major Aviation Accidents and Stakeholder Spillovers: A Comparative Case Study on Industry-Wide Crisis Transmission and Preparedness*

Richard Klophaus, Gerda Uebachs (Worms University of Applied Sciences, DE)

Synthesis

- Three modes represented: rail (P6), bus (P9), aviation (P18).
- P6 uses GAN-based synthetic data augmentation to improve railway delay prediction stability.
- P9 assesses circular electric bus network resilience, using entropy-based weighting.
- P18 examines stakeholder spillovers from major aviation accidents.

6. Traffic state and flow estimation

- **P1.** *Traffic State Transition Conditions refined for High-Resolution Aerial Observations*
Marc Zürn, Hubert Rehborn, Lennart Querfurth (Mercedes-Benz AG, DE); Robert Hoyer (University of Kassel, DE)
- **P3.** *Lane Distribution Characteristics across Traffic Phases using Probe Vehicle Data*
Daniel Arnold, Hubert Rehborn, Eric Grässer (Mercedes-Benz AG, DE); Jörg Eberhardt (Ravensburg-Weingarten University of Applied Sciences, DE)

Synthesis

- Both posters originate from the same Mercedes-Benz / academic collaboration.
- P1 refines traffic state transition conditions using high-resolution aerial observations.
- P3 characterizes lane distribution across traffic phases using probe vehicle data.

Closing

- Six themes covered: safety/equity/economics, freight and last-mile logistics, fleet electrification, digital platforms and AI tools, reliability and resilience, and traffic state estimation.
- Freight and logistics is the largest theme by poster count.
- Session 2P spans measurement, optimization, digital infrastructure, electrification, and safety/equity assessment in transport research.